

Kathan Sanghvi

(250) 883-1210 | Kelowna, BC | kathansanghvi5@gmail.com | [LinkedIn](#) | [Engineering Portfolio](#)

Summary

Engineer-in-Training with a mechanical engineering foundation and hands-on experience delivering electrical utility designs, offering a rare cross-disciplinary perspective on energy and infrastructure systems. Combines mechanical systems understanding with practical utility design experience to develop out-of-the-box solutions that are technically sound and economically feasible. Particularly motivated by complex energy systems challenges that demand strong technical judgment, coordination, and systems level thinking.

Education

Bachelor of Engineering – Mechanical | University of Victoria, BC

Sep 2020 – Apr 2025

Specializations: Energy Systems & Thermo-fluids/Aerodynamics

Engineering Skills & Experience

CAD & CAM	: AutoCAD, SolidWorks, Siemens NX; Mastercam, Vericut
Simulation & Analysis Software	: SPIDAcad, PV Elite, MATLAB, eQUEST, Ansys
Technical Drawings & Diagrams	: Single Line Diagrams, Process Flow Diagrams, P&IDs
Engineering Analysis	: FEA, CFD, Geometry Optimization, Sensitivity Analysis
Shop Tools	: CNC Mills, Manual Mills, Lathes, Laser Cutters, 3D Printers
Programming Language	: C, C++, G-Code, APT, R
Microsoft Office	: Excel, Access, Word, PowerPoint, Teams, SharePoint

Professional Experience

Electrical Designer | Primary Engineering and Construction | Kelowna, BC August 2025 – Present

- Drive full-cycle Customer Design Build projects from initial SLD through construction by ensuring inputs from developers, their consultants, and FortisBC are incorporated at each stage to deliver solutions that are technically and financially viable.
- Design overhead and underground distribution systems by performing detailed calculations, coordinating with stakeholders, minimizing costs, and producing construction ready drawings in AutoCAD.
- Secure utility approvals by preparing, submitting, and revising distribution designs to comply with FortisBC standards, the Canadian Electrical Code, and local government regulations.
- Support system improvement projects for FortisBC and distribution design projects for BC Hydro programs including Q3, Q9, and C3 EOL, demonstrating versatility across distinct utility standards.
- Assist project delivery by contributing to scope definition, schedule coordination, third-party service requests, and technical communication in support of timely and cost-aware execution.

Solar Panel Efficiency Research & Design Co-op | University of Victoria, BC

May – Aug 2024

Mentored by: [Dr. Andrew Rowe](#), Acting Director, [Institute for Integrated Energy Systems](#)

- Designed an autonomous solar panel cooling and cleaning system to reduce soiling and overheating losses in residential installations, improving energy yield and demonstrating technical and economic feasibility.
- Built a detailed SolidWorks model to validate system installation on a two-story house, using multiple design iterations to optimize flow paths, minimize pump load, and reduce water and energy waste.
- Engineered Excel and MATLAB models to predict panel surface temperatures, quantify system power requirements, and forecast annual savings, delivering clear cost-benefit insights to guide design decisions.

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Design Assistant Co-op | ZE Power Engineering | Richmond, BC May – Aug 2023, Jan – Aug 2022

- Executed design revisions on overhead and underground utility drawings, ensuring compliance with stringent BC Hydro standards and industry codes.
- Consolidated drawings, specifications, and supporting documents into field-ready job folders, coordinating with cross-functional teams to ensure alignment with site requirements and safety protocols.
- Coordinated with municipalities and private stakeholders to secure project permits and approvals, representing the design team in external discussions and helping navigate regulatory requirements.
- Prepared work orders and construction packages for a variety of BC Hydro projects, maintaining accuracy and seamless coordination using Excel, Access, and other Microsoft 365 tools.
- Conducted on-site evaluations across British Columbia, applying first principles to identify structural and environmental limitations and relaying practical design recommendations to the project team.

Academic Projects

Recirculating Water Bath | Capstone Project Sep 2024 – Apr 2025

Recipient of the Marianne Black Award for Best Engineering Design

- Developed a precision recirculating water bath with $\pm 0.1^{\circ}\text{C}$ tolerance for biomedical material testing, featuring variable heating, PI control, and a user-focused stainless-steel design.
- Conducted CFD analysis to optimize flow rate, pipe dimensions, and inlet/outlet geometry, achieving uniform thermal distribution and minimizing temperature gradients during prolonged operation.
- Gained hands-on experience in sheet metal work, plumbing, sensor integration, and control systems by fabricating 80-90% of components in-house, meeting stringent design constraints with limited resources.

Heat Transfer Coefficient Test Rig | Accent Refrigeration Systems Ltd. Jan – Apr 2025

- Designed a fully instrumented test rig for Accent Refrigeration to evaluate copper coil heat exchangers, integrating a 1 HP pump, 40 kW heater, and real-time flow, pressure, and temperature monitoring.
- Delivered comprehensive documentation, including P&ID schematics, equipment sizing charts, and CAD models, complemented by detailed testing guidelines to ensure consistent conditions.

Energy Efficient Building Design | HVAC Jan – Apr 2025

- Engineered HVAC systems by sizing ducts, diffusers, fans, and coils, integrating psychrometric and infiltration data with robust building envelope strategies to optimize thermal performance.
- Produced mechanical layout drawings using AutoCAD and performed energy simulations in eQUEST, applying several sustainability measures to reduce heating and cooling loads.

Volunteer Experience

Notetaker | [Centre for Accessible Learning](#), University of Victoria, BC Sep 2024 – Apr 2025

Instructor & Lunch Server | [Sneh Foundation](#), Vadodara, Gujarat, India Jul 2018 – Jan 2020

Additional Work Experience

Sales Associate (PT) | Cascadia Liquor Royal Bay, Victoria, BC Jun 2024 – July 2025

Design Assistant (PT) | ZE Power Engineering, Richmond, BC Jan – Apr 2024, Sep – Dec 2022

Associate (PT) | Hudson's Bay, Victoria, BC May – Aug 2022

Sales Associate (PT) | 7-Eleven, Victoria, BC Sep – Dec 2021